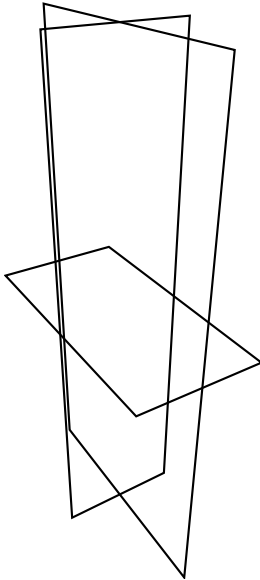


LINEAR ALGEBRA

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<http://joshua.smcvt.edu/linearalgebra>

Naturally we usually prefer representations that are easier to understand. We say that a map or matrix has been *diagonalized* when we find a basis B such that the representation is diagonal with respect to B, B , that is, with respect to the same starting basis as ending basis. Chapter Five finds which maps and matrices are diagonalizable.

The rest of this subsection develops the easier case of finding two bases B, D such that a representation is simple. Recall that the prior subsection shows that a matrix is a change of basis matrix if and only if it is nonsingular.

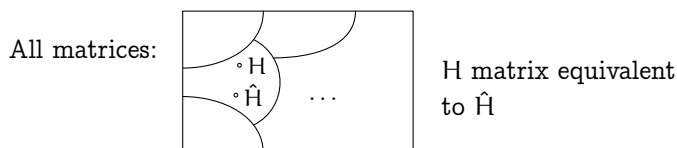
2.4 Definition Same-sized matrices H and $\hat{H}$ are *matrix equivalent* if there are nonsingular matrices P and Q such that $\hat{H} = PHQ$.

2.5 Corollary Matrix equivalent matrices represent the same map, with respect to appropriate pairs of bases.

PROOF This is immediate from equation $(*)$ above.

QED

Exercise 24 checks that matrix equivalence is an equivalence relation. Thus it partitions the set of matrices into matrix equivalence classes.



We can get insight into the classes by comparing matrix equivalence with row equivalence (remember that matrices are row equivalent when they can be reduced to each other by row operations). In $\hat{H} = PHQ$, the matrices P and Q are nonsingular and thus each is a product of elementary reduction matrices by Lemma IV.4.7. Left-multiplication by the reduction matrices making up P performs row operations. Right-multiplication by the reduction matrices making up Q performs column operations. Hence, matrix equivalence is a generalization of row equivalence—two matrices are row equivalent if one can be converted to the other by a sequence of row reduction steps, while two matrices are matrix equivalent if one can be converted to the other by a sequence of row reduction steps followed by a sequence of column reduction steps.

Consequently, if matrices are row equivalent then they are also matrix equivalent since we can take Q to be the identity matrix. The converse, however, does not hold: two matrices can be matrix equivalent but not row equivalent.

2.6 Example These two are matrix equivalent

$$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \quad \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$$

because the second reduces to the first by the column operation of taking -1 times the first column and adding to the second. They are not row equivalent because they have different reduced echelon forms (both are already in reduced form).

We close this section by giving a set of representatives for the matrix equivalence classes.

2.7 Theorem Any $m \times n$ matrix of rank k is matrix equivalent to the $m \times n$ matrix that is all zeros except that the first k diagonal entries are ones.

$$\begin{pmatrix} 1 & 0 & \dots & 0 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 & 0 & \dots & 0 \\ \vdots & & & & & & \\ 0 & 0 & \dots & 1 & 0 & \dots & 0 \\ 0 & 0 & \dots & 0 & 0 & \dots & 0 \\ \vdots & & & & & & \\ 0 & 0 & \dots & 0 & 0 & \dots & 0 \end{pmatrix}$$

This is a *block partial-identity* form.

$$\left(\begin{array}{c|c} I & Z \\ \hline Z & Z \end{array} \right)$$

PROOF Gauss-Jordan reduce the given matrix and combine all the row reduction matrices to make P . Then use the leading entries to do column reduction and finish by swapping the columns to put the leading ones on the diagonal. Combine the column reduction matrices into Q . QED

2.8 Example We illustrate the proof by finding P and Q for this matrix.

$$\begin{pmatrix} 1 & 2 & 1 & -1 \\ 0 & 0 & 1 & -1 \\ 2 & 4 & 2 & -2 \end{pmatrix}$$

First Gauss-Jordan row-reduce.

$$\begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 1 & -1 \\ 0 & 0 & 1 & -1 \\ 2 & 4 & 2 & -2 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 0 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Then column-reduce, which involves right-multiplication.

$$\begin{pmatrix} 1 & 2 & 0 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & -2 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Finish by swapping columns.

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

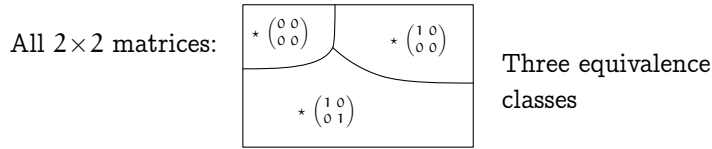
Finally, combine the left-multipliers together as P and the right-multipliers together as Q to get PHQ.

$$\begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 1 & -1 \\ 0 & 0 & 1 & -1 \\ 2 & 4 & 2 & -2 \end{pmatrix} \begin{pmatrix} 1 & 0 & -2 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

2.9 Corollary Matrix equivalence classes are characterized by rank: two same-sized matrices are matrix equivalent if and only if they have the same rank.

PROOF Two same-sized matrices with the same rank are equivalent to the same block partial-identity matrix. QED

2.10 Example The 2×2 matrices have only three possible ranks: zero, one, or two. Thus there are three matrix equivalence classes.



Each class consists of all of the 2×2 matrices with the same rank. There is only one rank zero matrix. The other two classes have infinitely many members; we've shown only the canonical representative.

One nice thing about the representative in Theorem 2.7 is that we can completely understand the linear map when it is expressed in this way: where the bases are $B = \langle \vec{\beta}_1, \dots, \vec{\beta}_n \rangle$ and $D = \langle \vec{\delta}_1, \dots, \vec{\delta}_m \rangle$ then the map's action is $c_1 \vec{\beta}_1 + \dots + c_k \vec{\beta}_k + c_{k+1} \vec{\beta}_{k+1} + \dots + c_n \vec{\beta}_n \mapsto c_1 \vec{\delta}_1 + \dots + c_k \vec{\delta}_k + \vec{0} + \dots + \vec{0}$ where k is the rank. Thus we can view any linear map as a projection.

$$\begin{pmatrix} c_1 \\ \vdots \\ c_k \\ c_{k+1} \\ \vdots \\ c_n \end{pmatrix}_B \mapsto \begin{pmatrix} c_1 \\ \vdots \\ c_k \\ 0 \\ \vdots \\ 0 \end{pmatrix}_D$$

Exercises

✓ 2.11 Decide if these are matrix equivalent.

(a) $\begin{pmatrix} 1 & 3 & 0 \\ 2 & 3 & 0 \end{pmatrix}, \begin{pmatrix} 2 & 2 & 1 \\ 0 & 5 & -1 \end{pmatrix}$

(b) $\begin{pmatrix} 0 & 3 \\ 1 & 1 \end{pmatrix}, \begin{pmatrix} 4 & 0 \\ 0 & 5 \end{pmatrix}$

(c) $\begin{pmatrix} 1 & 3 \\ 2 & 6 \end{pmatrix}, \begin{pmatrix} 1 & 3 \\ 2 & -6 \end{pmatrix}$

2.12 Which of these are matrix equivalent to each other?

(a) $\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$ (b) $\begin{pmatrix} 1 & 3 \\ -1 & -3 \end{pmatrix}$ (c) $\begin{pmatrix} -5 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix}$ (d) $\begin{pmatrix} 0 & -1 \\ 0 & 5 \end{pmatrix}$

(e) $\begin{pmatrix} 1 & 0 & 1 \\ 2 & 0 & 2 \\ 1 & 3 & 1 \end{pmatrix}$ (f) $\begin{pmatrix} 3 & 1 & 0 \\ 9 & 3 & 0 \\ -3 & -1 & 0 \end{pmatrix}$

✓ 2.13 Find the canonical representative of the matrix equivalence class of each matrix.

(a) $\begin{pmatrix} 2 & 1 & 0 \\ 4 & 2 & 0 \end{pmatrix}$ (b) $\begin{pmatrix} 0 & 1 & 0 & 2 \\ 1 & 1 & 0 & 4 \\ 3 & 3 & 3 & -1 \end{pmatrix}$

2.14 Suppose that, with respect to

$$B = \mathcal{E}_2 \quad D = \left\langle \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \end{pmatrix} \right\rangle$$

the transformation $t: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is represented by this matrix.

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$$

Use change of basis matrices to represent t with respect to each pair.

(a) $\hat{B} = \left\langle \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\rangle, \hat{D} = \left\langle \begin{pmatrix} -1 \\ 0 \end{pmatrix}, \begin{pmatrix} 2 \\ 1 \end{pmatrix} \right\rangle$

(b) $\hat{B} = \left\langle \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\rangle, \hat{D} = \left\langle \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 2 \\ 1 \end{pmatrix} \right\rangle$

2.15 What sizes are P and Q in the equation $\hat{H} = PHQ$?✓ 2.16 Consider the spaces $V = \mathcal{P}_2$ and $W = \mathcal{M}_{2 \times 2}$, with these bases.

$$B = \langle 1, 1+x, 1+x^2 \rangle \quad D = \left\langle \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \right\rangle$$

$$\hat{B} = \langle 1, x, x^2 \rangle \quad \hat{D} = \left\langle \begin{pmatrix} -1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & -1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \right\rangle$$

We will find P and Q to convert the representation of a map with respect to B, D to one with respect to $\hat{B}, \hat{D}$.

(a) Draw the appropriate arrow diagram.

(b) Compute P and Q .✓ 2.17 Find the change of basis matrices Q and P that will convert the representation of a $t: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ with respect to B, D to one with respect to $\hat{B}, \hat{D}$.

$$B = \left\langle \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\rangle \quad D = \left\langle \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \end{pmatrix} \right\rangle \quad \hat{B} = \mathcal{E}_2 \quad \hat{D} = \left\langle \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\rangle$$

- ✓ 2.18 Find the P and Q to express H via PHQ as a block partial identity matrix.

$$H = \begin{pmatrix} 2 & 1 & 1 \\ 3 & -1 & 0 \\ 1 & 3 & 2 \end{pmatrix}$$

- ✓ 2.19 Use Theorem 2.7 to show that a square matrix is nonsingular if and only if it is equivalent to an identity matrix.
- 2.20 Show that, where A is a nonsingular square matrix, if P and Q are nonsingular square matrices such that $PAQ = I$ then $QP = A^{-1}$.
- 2.21 Why does Theorem 2.7 not show that every matrix is diagonalizable (see Example 2.3)?
- 2.22 Must matrix equivalent matrices have matrix equivalent transposes?
- 2.23 What happens in Theorem 2.7 if $k = 0$?
- 2.24 Show that matrix equivalence is an equivalence relation.
- ✓ 2.25 Show that a zero matrix is alone in its matrix equivalence class. Are there other matrices like that?
- 2.26 What are the matrix equivalence classes of matrices of transformations on $\mathbb{R}^1$? $\mathbb{R}^3$?
- 2.27 How many matrix equivalence classes are there?
- 2.28 Are matrix equivalence classes closed under scalar multiplication? Addition?
- 2.29 Let $t: \mathbb{R}^n \rightarrow \mathbb{R}^n$ represented by T with respect to $\mathcal{E}_n, \mathcal{E}_n$.

- (a) Find $\text{Rep}_{B,B}(t)$ in this specific case.

$$T = \begin{pmatrix} 1 & 1 \\ 3 & -1 \end{pmatrix} \quad B = \left\langle \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} -1 \\ -1 \end{pmatrix} \right\rangle$$

- (b) Describe $\text{Rep}_{B,B}(t)$ in the general case where $B = \langle \vec{\beta}_1, \dots, \vec{\beta}_n \rangle$.

- 2.30 (a) Let V have bases B_1 and B_2 and suppose that W has the basis D . Where $h: V \rightarrow W$, find the formula that computes $\text{Rep}_{B_2,D}(h)$ from $\text{Rep}_{B_1,D}(h)$.

- (b) Repeat the prior question with one basis for V and two bases for W .

- 2.31 (a) If two matrices are matrix equivalent and invertible, must their inverses be matrix equivalent?

- (b) If two matrices have matrix equivalent inverses, must the two be matrix equivalent?

- (c) If two matrices are square and matrix equivalent, must their squares be matrix equivalent?

- (d) If two matrices are square and have matrix equivalent squares, must they be matrix equivalent?

- 2.32 Square matrices are *similar* if they represent the same transformation, but each with respect to the same ending as starting basis. That is, $\text{Rep}_{B_1,B_1}(t)$ is similar to $\text{Rep}_{B_2,B_2}(t)$.

- (a) Give a definition of matrix similarity like that of Definition 2.4.

- (b) Prove that similar matrices are matrix equivalent.

- (c) Show that similarity is an equivalence relation.

- (d) Show that if T is similar to $\hat{T}$ then T^2 is similar to $\hat{T}^2$, the cubes are similar, etc.

- (e) Prove that there are matrix equivalent matrices that are not similar.